Blessed G/Michael Catholic Secondary School

Semester 1 Mathematics Mid1 Examination for Grade11

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade11 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** What is the domain of the rational function f(x) = (2x + 1) / (x^2 - 9)?

A. All real numbers except x = 3
B. All real numbers except x = -3
C. All real numbers except x = 3 and x = -3
D. All positive real numbers

____ **2.** What is the domain of the rational function f(x) = (2x + 1) / (x^2 - 9)?

A. {x in R | x != 3 and x != -3} **B.** {x in R | x != 9}
C. {x in R | x > 3} **D.** All real numbers R

____ **3.** What is the horizontal asymptote of the rational function f(x) = (3x^2 - 5) / (2x^2 + 7x - 4)?

A. y = 0 **B.** y = 3/2
C. y = -5/4 **D.** No horizontal asymptote

____ **4.** What are the coordinates of the center and radius of the circle with equation (x - 4)^2 + (y + 2)^2 = 25?

A. Center (4, -2), radius = 5 **B.** Center (-4, 2), radius = 5
C. Center (4, -2), radius = 25 **D.** Center (-4, 2), radius = 25

____ **5.** What is the slope of a line perpendicular to 2x - 5y + 10 = 0?

A. 2/5 **B.** -2/5 **C.** 5/2 **D.** -5/2

____ **6.** Which of the following functions is an odd function?

A. f(x) = x^2 + 4 **B.** f(x) = x^3 - 5x
C. f(x) = |x| **D.** f(x) = cos(x)

____ **7.** Solve for x: (x + 2) / (x - 1) = 3

A. x = 2.5 **B.** x = 3 **C.** x = 1.5 **D.** x = 4

____ **8.** Decompose 5 / [(x - 1)(x + 4)] into partial fractions: A/(x - 1) + B/(x + 4). What is the value of A?

A. 1 **B.** 2 **C.** -1 **D.** 5

____ **9.** What is the distance between the points P(1, 2) and Q(4, 6)?

A. 3 **B.** 4 **C.** 5 **D.** 7

____ **10.** Find the vertex of the parabola y = x^2 - 4x + 7:

A. (2, 3) **B.** (-2, 3) **C.** (2, -3) **D.** (4, 7)

____ **11.** Which of the following is a pure scalar physical quantity?

A. Displacement **B.** Linear momentum
C. Kinetic energy **D.** Gravitational force

____ **12.** Two vectors A and B have magnitudes of 6 units and 8 units respectively. If they are perpendicular (at 90°), what is the magnitude of their resultant vector R = A + B?

A. 10 units **B.** 14 units **C.** 2 units **D.** 48 units

____ **13.** A projectile is launched from ground level with initial velocity v0 at an angle θ above the horizontal. At the peak of its trajectory, which of the following is TRUE?

A. Both vertical and horizontal velocity components are zero
B. Vertical velocity vy is zero, while horizontal velocity vx = v0 cos(θ)
C. Acceleration is zero
D. Vertical velocity vy = v0 sin(θ)

____ **14.** A 5 kg block rests on a horizontal frictionless surface. A horizontal force of 20 N is applied to it. What is the acceleration of the block?

A. 2 m/s² **B.** 4 m/s² **C.** 10 m/s² **D.** 100 m/s²

____ **15.** What work is done by a constant force of 15 N pushing an object through a displacement of 4 meters in the direction of the force?

A. 30 Joules **B.** 60 Joules **C.** 3.75 Joules **D.** 19 Joules

____ **16.** According to the work-energy theorem, the net work done by all forces acting on a particle equals the change in its:

A. Kinetic energy **B.** Potential energy

C. Total momentum **D.** Rest mass

____ **17.** A uniform circular motion at constant speed v in a circle of radius r has a centripetal acceleration directed toward:

A. The center of the circular path
B. Tangent to the circle in the direction of motion
C. Radially outward from the circle
D. Opposite to the velocity vector

____ **18.** In an isolated system where only conservative forces do work, the total mechanical energy (E = K + U) is:

A. Constantly increasing **B.** Conserved and remains constant

C. Zero at all times **D.** Equal to net friction

____ **19.** The rotational analog of linear mass (inertia) in rotational dynamics is:

A. Torque **B.** Moment of inertia (rotational inertia)

C. Angular velocity **D.** Centrifugal force

____ **20.** A torque τ = 40 N·m acts on a rigid body with moment of inertia I = 8 kg·m². What is the angular acceleration α?

A. 5 rad/s² **B.** 0.2 rad/s² **C.** 320 rad/s² **D.** 48 rad/s²

____ **21.** An elastic collision between two isolated bodies is characterized by the conservation of:

A. Both total linear momentum and total kinetic energy
B. Linear momentum only, while kinetic energy is transformed into heat
C. Kinetic energy only, but not linear momentum
D. Mechanical potential energy only

____ **22.** Newton's Universal Law of Gravitation states that the gravitational attractive force between two point masses m1 and m2 separated by distance r is proportional to:

A. r² **B.** 1 / r² **C.** 1 / r **D.** r

____ **23.** What power is expended when a machine lifts a 200 kg mass vertically through a height of 10 meters in 5 seconds? (Take g = 9.8 m/s²)

A. 1960 W **B.** 3920 W **C.** 980 W **D.** 490 W

____ **24.** When a skater pulls in her arms while spinning, her spin rate increases. This phenomenon is a direct consequence of:

A. Conservation of angular momentum **B.** Conservation of linear kinetic energy

C. External gravitational torque **D.** Increase in moment of inertia

____ **25.** A spring with spring constant k = 200 N/m is compressed by 0.1 m from its equilibrium length. What elastic potential energy is stored in the spring?

A. 1 Joule **B.** 2 Joules **C.** 10 Joules **D.** 20 Joules

____ **26.** What is the vertical asymptote of the rational function f(x) = (x + 3) / (2x - 8)?

A. x = 4 **B.** x = -4 **C.** x = 1/2 **D.** x = -3

____ **27.** Which of the following functions has an oblique (slant) asymptote?

A. f(x) = (x^2 - 1) / (x + 2) **B.** f(x) = (x - 1) / (x^2 + 2)

C. f(x) = (3x^2 + 1) / (x^2 - 4) **D.** f(x) = 1 / x

____ **28.** What is the y-intercept of the rational function f(x) = (x^2 - 4x + 6) / (x - 2)?
A. (0, -3) **B.** (0, 3) **C.** (0, -2) **D.** (0, 6)

____ **29.** What is the solution set of the rational inequality (x - 2) / (x + 3) <= 0?
A. (-3, 2] **B.** [-3, 2]
C. (-infinity, -3) U [2, infinity) **D.** (-3, 2)

____ **30.** Find the radius of the circle given by x^2 + y^2 - 6x + 8y = 0:
A. 5 **B.** 10 **C.** 25 **D.** 7

____ **31.** What is the equation of the line passing through (2, 3) with slope m = -4?
A. 4x + y - 11 = 0 **B.** 4x - y + 5 = 0
C. 4x + y + 11 = 0 **D.** x + 4y - 14 = 0

____ **32.** Two lines L1: 3x - 2y = 5 and L2: 6x - 4y = 12 are:
A. Perpendicular **B.** Parallel and distinct
C. Coincident **D.** Intersecting at one point

____ **33.** What is the midpoint of the line segment joining A(-3, 7) and B(5, -1)?
A. (1, 3) **B.** (2, 6) **C.** (1, 4) **D.** (-1, 3)

____ **34.** What is the range of the rational function f(x) = (2x - 1) / (x + 3)?
A. {y in R | y != 2} **B.** {y in R | y != -3}
C. {y in R | y != 1/2} **D.** All real numbers R

____ **35.** If f(x) = 1 / x, what is the average rate of change of f(x) on the interval [1, 4]?
A. -1/4 **B.** -3/4 **C.** 1/4 **D.** -1/12

____ **36.** Solve the equation: 2 / x + 3 / (x + 1) = 0
A. x = -2/5 **B.** x = 2/5 **C.** x = -5/2 **D.** x = -1

____ **37.** Which of the following is an equation of a circle tangent to the x-axis with center (3, 4)?
A. (x - 3)^2 + (y - 4)^2 = 16 **B.** (x - 3)^2 + (y - 4)^2 = 9
C. (x - 3)^2 + (y - 4)^2 = 25 **D.** (x + 3)^2 + (y + 4)^2 = 16

____ **38.** If the rational expression (3x + 11) / (x^2 + 5x + 6) is decomposed into A/(x + 2) + B/(x + 3), find the value of B:
A. -2 **B.** 5 **C.** 3 **D.** -1

____ **39.** The distance from point P(2, 1) to the line 3x + 4y - 5 = 0 is:
A. 1 unit **B.** 2 units **C.** 5 units **D.** 0.5 units

____ **40.** What is the domain of the function f(x) = sqrt((x - 2) / (x + 5))?
A. (-infinity, -5) U [2, infinity) **B.** (-5, 2]
C. [-5, 2] **D.** [2, infinity)